

OAKLEY 19 SSW, KS, USA

WMO: 724570

Lat: 38.8701N

Lon: 100.9627W

Elev: 2870

StdP: 13.23

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 03067

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	1.4	6.4	-6.4	4.3	5.3	-2.0	5.5	12.1	26.3	33.7	23.7	30.1	6.1	N/A	0.537

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.4	99.9	67.5	96.9	67.7	94.0	67.4	72.5	88.2	71.2	87.5	70.2	86.7	10.1	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.4	116.0	77.4	66.9	110.0	76.5	65.6	105.1	75.6	38.1	88.6	36.9	87.7	35.9	86.8	76.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
23.6	20.3	17.4	-7.7	105.2	5.4	3.1	-11.6	107.4	-14.7	109.2	-17.7	110.9	-21.7	113.1	
			DB	-9.3	74.7	5.6	1.4	-13.3	75.6	-16.6	76.4	-19.8	77.2	-23.9	78.2
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.8	30.8	32.3	44.0	51.9	61.7	74.0	78.5	76.0	68.4	53.8	41.8	30.4	
	DBStd	19.48	10.49	11.57	10.64	9.42	8.93	6.85	5.48	5.30	8.19	9.75	9.61	10.56	
	HDD50	2397	595	502	241	89	13	0	0	0	1	72	277	607	
	HDD65	5395	1059	916	652	401	169	9	1	1	58	363	695	1071	
	CDD50	3768	1	6	55	144	376	720	885	806	552	190	32	1	
	CDD65	1290	0	0	1	6	67	279	421	342	159	15	0	0	
	CDH74	16063	2	15	104	274	1043	3459	5083	3750	1914	387	32	0	
	CDH80	8114	0	3	22	73	410	1787	2841	1951	900	125	2	0	
Wind	WSAvg	7.5	6.9	7.6	8.5	9.8	8.6	7.8	6.8	6.6	7.0	7.3	6.9	6.7	
Precipitation	PrecAvg	19.3	0.4	0.4	1.0	1.9	2.8	2.5	3.5	2.6	1.6	1.4	0.7	0.5	
	PrecMax	25.9	1.9	1.6	3.1	4.4	6.3	4.9	8.4	5.2	4.4	4.0	2.7	3.9	
	PrecMin	12.6	0.0	0.0	0.0	0.6	1.3	0.5	0.2	0.5	0.3	0.0	0.0	0.0	
	PrecStd	3.3	0.5	0.4	0.7	0.9	1.4	1.2	1.9	1.3	1.1	1.3	0.7	0.8	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	68.3	75.7	83.2	87.2	95.3	105.5	102.9	101.8	98.5	89.2	78.2	65.3	
		MCWB	46.6	49.6	54.6	55.1	60.6	64.9	68.9	67.4	64.9	60.6	52.8	45.8	
	2%	DB	60.1	66.3	76.6	81.5	88.9	98.4	100.0	97.5	93.9	83.4	72.5	59.0	
		MCWB	43.2	46.2	52.3	55.5	62.8	66.4	68.6	68.5	65.1	58.0	51.0	42.6	
	5%	DB	54.2	58.6	70.3	76.6	84.6	94.3	97.3	94.3	89.9	78.2	65.6	53.3	
		MCWB	40.7	42.8	49.5	54.4	61.5	67.1	69.1	67.9	64.9	56.2	47.6	41.4	
	10%	DB	47.7	51.7	64.0	70.6	80.0	90.2	94.2	90.7	85.3	72.3	59.2	47.2	
		MCWB	37.5	39.6	47.7	52.7	60.5	66.9	68.9	67.8	64.1	54.0	45.2	38.1	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.7	50.4	59.7	61.2	69.2	73.1	74.5	73.8	70.8	64.6	55.3	49.2
			MCDB	63.8	68.6	73.2	75.8	82.8	87.7	90.0	88.5	86.9	76.3	67.0	56.4
2%		WB	43.7	46.8	54.5	58.9	66.5	71.2	72.8	72.0	68.7	61.0	52.3	44.0	
		MCDB	58.1	66.2	69.2	73.0	79.1	87.4	88.8	87.5	84.5	75.5	66.4	55.7	
5%		WB	40.8	43.6	51.5	56.6	64.4	69.8	71.5	70.8	67.4	58.3	49.2	41.0	
		MCDB	52.7	58.4	67.3	69.9	78.0	86.5	87.9	86.5	83.1	73.4	64.7	53.0	
10%		WB	37.9	40.0	48.5	54.2	62.3	68.4	70.4	69.6	66.0	55.5	45.9	37.8	
		MCDB	46.7	51.4	64.6	68.3	76.6	84.8	87.6	85.1	80.8	68.3	59.1	47.2	
Mean Daily Temperature Range	5% DB	MDBR	26.6	26.8	29.6	27.9	26.8	27.5	26.4	26.1	27.3	28.5	29.3	25.5	
		MCDBR	37.3	39.9	40.4	37.7	33.5	32.8	31.0	31.6	32.6	37.6	39.1	35.7	
		MCWBR	22.0	21.9	20.0	16.9	13.4	10.1	8.5	9.9	10.8	16.5	20.6	21.8	
	5% WB	MCDBR	34.4	39.0	36.9	30.3	28.5	28.5	27.3	26.6	27.3	31.3	35.8	32.8	
		MCWBR	21.1	22.3	19.5	15.5	13.5	11.3	9.0	9.6	10.0	15.3	19.1	20.5	
Clear-Sky Solar Irradiance	taub	0.260	0.267	0.297	0.324	0.351	0.371	0.391	0.368	0.340	0.296	0.267	0.257		
	taud	2.545	2.547	2.469	2.419	2.411	2.388	2.376	2.402	2.459	2.564	2.567	2.552		
	Ebn at Noon	298	309	307	302	294	287	280	285	288	294	292	291		
	Edh at Noon	23	26	32	36	37	38	38	36	32	26	23	21		
All-Sky Solar Radiation	RadAvg	813	1073	1477	1810	2094	2321	2299	1998	1656	1213	895	713		
	RadStd	65	98	125	97	129	111	81	110	96	122	69	49		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page